TROUBLESHOOTING ASSISTANT, PROOF OF CONCEPT

FixPath

Every answer shows its source and the graph path behind it.

Customer: a fictional Tier-1 automotive supplier. Line 3, Singapore. Five machines, night shift.

A discovery plan, an MVP, and a working proof of concept.

Five places to look at a stopped machine

MACHINE LOGS

5

files, four date formats

HANDOVER NOTES

90

notes, shorthand and typos

SPARES SHEETS

2

lists that disagree

MAINTENANCE NOTES

6

files, free text

PDF MANUALS

4

vendor manuals

ASSUMPTION Nobody has measured the time from "machine down" to the first correct action. The workshop sets that baseline: a stopwatch on ten real breakdowns.

A half-day workshop on site

WHO ATTENDS

Maintenance lead. Two or three technicians, one from nights. Production supervisor. IT and OT engineer. Quality engineer. EHS representative.

BLOCK	OUTPUT
Last five breakdowns	Map of where the time went
Questions and sources	Top 20 questions, source map
Open the folder together	Data inventory
Slice the MVP	One line, one persona, top failure modes
Success and limits	Success metrics, constraint list

Questions that narrow the scope

PROBLEM

Which three machines cause the most downtime?

Where does breakdown time go: diagnosis, finding information, parts or approvals?

USERS

Who troubleshoots at night, at what skill level?

What device do they carry? Do gloves, noise or language get in the way?

DATA

Which source do technicians trust?

How current is the spares sheet? Who owns each source?

CONSTRAINTS

Are cloud models allowed, or only on-prem? What must the assistant never say about safety? Are IT and OT networks separated?

SUCCESS

Does "fixed faster" mean MTTR, time to first correct action or first-time fix rate? Is there a baseline, and who signs off the pilot?

MAKE ASSUMPTIONS VISIBLE

Assumptions, and how the workshop checks each

ASSUMPTION	HOW THE WORKSHOP VALIDATES IT
The shop floor works in English, in Singapore	Ask directly, check the language of existing notes
The MVP covers one line with five machines	Pareto chart of downtime by machine
Technicians share a tablet on the floor	Observe one shift
Data arrives as a shared-folder export, with no MES, CMMS or historian access	IT and OT inventory session
Cloud models are allowed for the pilot	IT security sign-off
The assistant is read-only and writes no work orders	Agree it in the MVP scope
Lockout and tagout are mandatory for hydraulic and electrical work	EHS representative confirms the procedure list
The success metric is time to the first correct action	Stopwatch baseline on ten real breakdowns

MVP: one line, one persona, read-only

IN

One line, five machines

One persona: the night-shift technician

Likely cause, fix steps, parts with stock and bin, similar past breakdowns, who fixed it last

Every answer cites its source

Lockout block, always shown when a fix needs it

OUT

Writing work orders

MES and CMMS integration

Photo input from HMI screens

On-prem models

Other lines

SUCCESS

Time to the first correct action, against the stopwatch baseline

Lockout recall of 100%

Thumbs up or down on each answer, stored as a score

Five sources, and what is wrong with each

SOURCE	FILES	A REAL LINE FROM THE FOLDER

Machine logs	5	`2026-04-01T04:00:00Z HB ok`
		`01/04/2026 06:45:00 | ERR 200 | PUMP START`
		`1774994594;RB-551;GRIPPER PART-PRESENT SIGNAL LOST`

Shift handovers	90	`leak test stopped 0631 w/ VT-012, good parts failing leak test,` `reference off. Recalibrate pressure senosr, tested ok.`
Spares sheets	2	`BRG7014ACD;Spindle angular contact;4;A04` `SEAL-KIT-50 | Main cylinder seal kit, 50 mm rod | 3 | pc | B12`
Maintenance notes	6	`2026-04-14,PR-2,"press losing pressure mid-stroke, oil on the floor"` `17/04/2026,Kessler 1,"ALM CL-210: coolant pressure dropping"`
PDF manuals	4	`Plant ref: PROC-RW-01 | Lockout required: yes | Time: 40 min`

ONTOLOGY

Nine node types, twelve links, five colour families

Every link reads as a sentence. A line has machines. A machine raised an alarm. An alarm usually means a root cause. A root cause is fixed by a fix. A fix needs a spare part. A past breakdown happened on a machine, started with an alarm, was caused by a root cause and used a part. Source backs each one.

One handover line becomes graph facts

INC-2026-016 is a past breakdown.

It **happened on** LT-01.

It **started with** VT-012.

It **was caused by** pressure sensor drift. The linker (JEV) picked that cause.

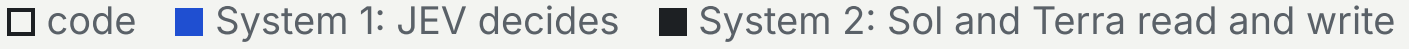
Evidence: the highlighted line, read by the text reader at 0.80. The maintenance export row for the same breakdown backs it at 0.90, so two sources merged into one node.

The ingestion crew



107 files, 164 nodes, 442 links, 906 evidence quotes, 37 items in the review queue.

The troubleshooting crew



One question, end to end

VALIDATION

What holds, and what it rests on

LOCKOUT RECALL

19 of 19

fixes that need lockout, all warned

ROOT CAUSE, TOP 1

20 of 20

questions with a known cause

TRAPS PASSED

5 of 5

no Line 4, vague, off topic, conflicting notes, zero stock

TRIAGE, ACTED ON

22 of 22

all five fields right; 29 of 32 on all 32 questions

CITATION PRECISION

96 of 102

kept claims judged supported (0.94)

ONE LIVE ANSWER

$0.01

median cost, 13.6 s median, n = 25

LIMITS

- The 25 golden questions re-ask recorded incidents, so these are not hold-out results.
- Over-warning (1 of 1) and the safety false-positive rate (0 of 1) each rest on one negative case.
- Live latency and cost were measured before the last code-only fixes.

JEV and Decisions: comparable accuracy

TASK	N	JEV	DECISIONS
Pick the root cause	60	100%	100%
Same part?	41	100%	100%
Same incident?	111	98%	96%
Triage fields	32	98%	98%
Verify a claim	102	95%	93%

Each backend at its own best gate. No task is led by either backend (sign test). 0 led for JEV, 0 for Decisions, 5 ties.

COST PER 1,000 ASKS

$0.04 vs $0.07

JEV vs Decisions, 266 asks each

TRIAGE LATENCY, P50

0.27 s vs 0.41 s

p95: 0.39 s vs 0.61 s, n = 30 asks

LIMITS

- The gates and prompts were tuned on JEV. The best gates are picked on the same questions they are scored on, so they are in sample.
- One LLM judge labels the verify claims. Latency is one run of 30 triage asks per backend, not interleaved.

NEXT

Roadmap, risks and how we keep validating

RISKS AND MITIGATIONS

RISK	MITIGATION
A model invents a fix step	Cited claims, JEV verifier, citation precision measured
A wrong merge joins two machines	Confidence gates, review queue
A safety warning goes missing	Code builds the block, recall must stay 100%
Stock data is stale	Shows the stock date, flags sources that disagree
A decider vendor fails	Second backend, replay mode

AFTER THE PROOF OF CONCEPT

Shadow-mode pilot on real breakdowns

Time to first correct action against the stopwatch baseline

Thumbs up or down on every answer, stored as a score

Expert review of the merge queue

ROADMAP

Graph database at scale. Asset Administration Shell mapping. CMMS and MES integration. Work orders written back. Photos from HMI screens. On-prem models.

NEXT

Maintenance contracts and SLAs

Link each machine to its contract terms: response time, uptime target and the parts covered.

Measure uptime against the target from the graph, which already records each breakdown's date and downtime.

Measure response time from when the alarm was raised to when the technician arrived.

Send the customer a monthly service report where every number links back to a recorded breakdown.

Warn before a breach: use the upkeep predictions to flag a part that is due to fail on a machine whose uptime is near its target.

Demo: ten minutes, live

1. Ask: HP-01 stopped with E-217
2. The path lights up, and a magnet opens its source
3. The lockout block
4. Line 4: the clarify
5. The zero-stock gripper
6. Pipeline: drop a new handover
7. Evals: scores and limits

Runs with no keys too: `make api-replay` and `make web`. The repository README has setup, the data and the eval summary.

Schema detail: what is in the graph

TYPE	LABEL	GLYPH	FAMILY	IN THE GRAPH
Line	Production line	factory	Where	1
Asset	Machine	cog	Where	5
Component	Component	bolt	Where	15
AlarmCode	Alarm	siren	Problem	20
FailureMode	Root cause	unlink	Problem	11
Procedure	Fix	wrench	Fix	12
Part	Spare part	package	Fix	39
Incident	Past breakdown	history	History	61
Document	Source	file-text	Source	107

Links in the graph: 442. Largest groups: used 105, caused by 61, happened on 61, started with 54, fits 51.

Decider questions and gates

ROLE	QUESTION TO JEV	GATE
Triage	Five in one ask: machine, alarm, root cause, intent, safety	Act from 0.75, confirm chips from 0.40
Merge a part or machine	Same entity?	Merge from 0.70, review from 0.50
Merge a breakdown	Same incident?	Merge from 0.90, review from 0.50, veto on a contradicting answer
Link	Which root cause for this breakdown?	Link from 0.75, review from 0.40
Verify	Does the quote support this claim?	Keep when supported plus partly reaches 0.50

The gates are tuned on eval layers 2 and 3. For same entity, every true pair scored 0.72 or more and the highest wrong pair 0.44.

Cost model

ITEM	MEASURED
Whole project, every call booked	$0.98 in the spend ledger
Last full ingest of 107 files	$0.0026, 83 paid calls of 302
One live answer	$0.0098 median, $0.0114 p95 (n = 25)
Decider, per 1,000 asks	JEV $0.040, Decisions $0.070
Hard stop	$40, set in the spend ledger

Everything replays from saved answers, so a rerun costs nothing. Live cost was measured before the last code-only fixes.

BACKUP

Failure handling

WHAT GOES WRONG	WHAT THE SYSTEM DOES
Not sure which machine, alarm or cause	Confirm with chips, or ask back. Never guess
A place the plant does not have (Line 4)	Says so and offers the closest machine
Not a maintenance question	Says so and names who to ask
A claim the quote does not support	Moves to unverified, never shown on the card
A part with zero stock	Shows "0 on hand" with the lead time
Two notes disagree	Goes to the review queue as a conflict
A model step fails	The card still shows the graph facts and a note
Spend reaches the budget	Hard stop, no further call starts
A decider API is down	Second backend, or replay mode for the demo